

Beatrice BATTISTI

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EDUCATION

Politecnico di Torino & Université de Bordeaux <i>Doctor of Philosophy, Applied Mathematics and Mechanical Engineering</i> <i>Supervisors: Dr. Michel Bergmann, Prof. Giovanni Bracco</i> <i>Defense: 22nd April 2024</i>	Nov 2020 – Apr 2024 <i>Cotutelle programme, with high honours</i>
Université Claude Bernard - Lyon 1 & Ecole Centrale de Lyon <i>Master of Science, Fluid Mechanics and Energy</i>	Sep 2017 – Oct 2018
Ecole Centrale de Lyon <i>Master of Science, General Engineering</i> <i>Major in Aeronautics</i>	Sep 2016 – Oct 2018 <i>Double Degree</i>
Politecnico di Torino <i>Master of Science, Aerospace Engineering and Astronautics</i> <i>Major in Aerodynamics</i>	Sep 2015 – Oct 2018 <i>Double Degree, with high honours</i>
Politecnico di Torino <i>Bachelor of Science, Aerospace Engineering</i>	Sep 2012 – Sep 2015 <i>with high honours</i>

RESEARCH INTERESTS

I am interested in developing numerical schemes for partial differential equations to describe the physics of environmental phenomena. My focus is on the numerical modeling of multi-phase flows and the implementation of model order reduction methods for non-linear problems.

CURRENT POSITION

CNRS iMPT, Université Savoie Mont Blanc <i>Post-doctoral Fellow</i> <i>ERUPTA: Efficient and Reliable numerical methods to Unravel Processes in magma Transport at Active volcanoes</i>	Sep 2025 – present <i>Supervisors: Walter Boscheri, Alain Burgisser</i>
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PROFESSIONAL AND RESEARCH EXPERIENCE

CNRS, Université Savoie Mont Blanc (LAMA) <i>Post-doctoral fellowship - Supervisor: Walter Boscheri</i> Numerical simulation and analysis of multiphase flows applied to volcanic phenomena	Chambéry, France Sep 2024 – Aug 2025
..... After PhD Before PhD	
Politecnico di Torino <i>Pre-Doctoral program</i> Wave energy converter array development	Turin, Italy Apr 2020 – Oct 2020
Politecnico di Torino <i>Research scholarship</i> Development of a CFD in-house code - Application to the wave energy	Turin, Italy Jan 2020 – Apr 2020
IKOS Consulting, at ALSTOM <i>Engineering full-time job</i> Numerical simulation of train dynamics in extreme conditions - Application to trams and to trains for the "TGV 2020" project	La Rochelle, France Mar 2019 – Dec 2019

INRIA Bordeaux - Sud-Ouest

Internship

- Numerical simulation of undulating bores in open channels and study of the influence of the sloping banks on the flow dynamics

Naval Group

Internship

- Hydrodynamics of the towed "V-Wing" systems and dimensioning of the drop cable

Bordeaux, France

Apr 2018 – Sep 2018

Ollioules, France

Apr 2017 – Aug 2017

RESEARCH STAYS

Optimad Srl

Secondment, ARIA (Accurate ROMs for Industrial Applications) project

- Body-less approach for the numerical modeling of wave energy converters

Ghent University

WECANet Short Term Scientific Mission (STSM)

- Numerical modeling of the far-field effects of a PeWEC farm

CIRM

Research project in the context of the summer school CEMRACS 2021

- Model Order Reduction of one-dimensional non-linear transport PDEs in porous media

Turin, Italy

Winter 2023 – 3 months

Ghent, Belgium

Fall 2022 – 3 weeks

Luminy, France

Summer 2021 – 5 weeks

AWARDS, DISTINCTIONS AND RESEARCH GRANTS

- MSCA Seal of Excellence, Marie Skłodowska-Curie Actions Postdoctoral Fellowships (European Commission). Project: *LIMEN: Leading Innovation in Multiphase Environmental Numerics*. Final score: 89.60/100. Winter 2026.
- WECANet COST Action Short-Term Scientific Mission (STSM) Fellowship, Ghent University - Numerical modeling of WEC farms. 4,000€. Fall 2022.
- CEMRACS Summer School Fellowship, CIRM - Reduced-order modeling of transport PDEs in porous media. Full support. Summer 2021.

ORGANIZATION OF SCIENTIFIC MEETINGS

Université Savoie Mont Blanc

Co-organizer of the Girls and Maths day

- with Laure Bastide, Georges Comte and Céline Labart (LAMA Equality Committee)

LAMA - Université Savoie Mont Blanc

Organizer of the PhD and Postdoc days in Mathematics of USMB

Université Savoie Mont Blanc

Co-organizer of the 3C conference

- 3C: Challenges in Computational methods for Complex environmental applications
- with Laure Bastide, Walter Boscheri and Arnaud Duran

LAMA - Université Savoie Mont Blanc

Co-organizer of the PhD and Postdoc day in Mathematics of USMB

- with Cassandre Lebot

Chambéry, France

Upcoming - May 2026

Yenne, France

October 2025

Chambéry, France

May 2025

Chambéry, France

November 2024

PUBLICATIONS

Summary (as of April 2026):

Google Scholar: 16 publications, 71 citations, h -index = 4.

Scopus: 10 publications, 28 citations, h -index = 3.

The complete list of publications is provided in Annex 1.

Dissertations

- Multi-fidelity multi-scale numerical modeling of wave energy converter farms. Doctoral thesis, 2024.
- Modeling of the upstream propagation of coastal surface waves into open channels - Application to the dynamics of undulating bores. Master thesis, 2018.
- Implementation of an auxiliary instrument for the design of a wingsail. Bachelor thesis, 2015.

TEACHING ACTIVITIES

2025/2026 – 1st semester (45 hours)

Université Savoie Mont Blanc

Chambéry, France

Course "Partial differential equations and discretization"

~ 15 students

- Supervision and evaluation of the practical sessions of the course (using Python), for 1st year master's students.
Main topics: finite differences, finite volumes, finite elements, stability and convergence.

Course "Mathematical modeling and scientific computing"

~ 15 students

- Supervision and evaluation of the practical sessions of the course (using Python), for 1st year master's students.
Main topics: ordinary differential equations, dynamical systems, parametric analysis and numerical errors.

Course "Numerical analysis"

< 10 students

- Supervision and teaching of exercise sessions of the course, for 3rd year bachelor's students.
Main topics: linear/nonlinear systems, numerical interpolation, numerical derivation and integration.

Course "Discrete probability"

~ 25 students

- Supervision and teaching of exercise sessions of the course, for 2nd year bachelor's students.
Main topics: combinatorics, sets and relations, conditional probability, probabilistic models, and discrete random variables.

2023/2024 – 1st semester (48 hours)

Enseirb-Matmeca

Bordeaux, France

Course "Algorithm and Programming in Fortran 90"

~ 15 students

- Supervision and evaluation of the practical sessions of the course, for 3rd year bachelor's students.
Main topics: programming fundamentals, quadrature, interpolation, linear systems, ODEs.

2022/2023 – 2nd semester (20 hours)

Enseirb-Matmeca

Bordeaux, France

Course "Algorithm and Programming in Fortran 90"

~ 15 students

- Supervision and evaluation of the practical sessions of the course, for 3rd year bachelor's students.
Main topics: finite differences for Poisson, heat, advection, and wave propagation equations.

PRESENTATIONS

Invited Talks Typically ≥ 30 min presentation; full support provided (conference fee and accommodation covered).

- Structure-preserving numerical schemes for complex multiscale flows. MACS Seminar at IMAG, Montpellier, France - Mar 2026.
- Numerical schemes for complex multi-scale flows. AMAC Seminar at LJK, Saint-Martin-d'Hères, France - Feb 2026.
- A numerical scheme for compressible two-phase flows at all Mach numbers. ETNA Conference, Catania, Italy - Nov 2025.
- A numerical scheme for compressible two-phase flows at all Mach numbers. JEARA Conference, Lyon, France - Nov 2025.
- A numerical scheme for compressible two-phase flows at all Mach numbers. EDP Seminar at LAMA, Le Bourget-du-Lac, France - Sep 2025.
- Multi-fidelity multi-scale numerical modeling of wave energy converter farms. INORE Symposium, Zarautz, Spain - Oct 2022.

Contributed Talks

- A multi-fidelity coupling methodology for the simulation of wave energy converters. Digital Ocean for the Future, Inria Webinar Series, online - Jan 2026.
- A numerical scheme for compressible two-phase flows at all Mach numbers. ARIA Workshop, Bidart, France - Sep 2025.
- A numerical scheme for compressible two-phase flows at all Mach numbers. Mathematics of Compressible Fluids Workshop, Clausthal, Germany - Jun 2025.

- A numerical scheme for compressible two-phase flows at all Mach numbers. Shark-FV Conference, Porto, Portugal - Jun 2025.
- A numerical scheme for compressible two-phase flows at all Mach numbers. 3C Conference, Chambéry, France - May 2025.
- Multi-query analysis of a PeWEC farm. EWTEC Conference, Bilbao, Spain - Sep 2023.
- A multi-fidelity coupling methodology for the simulation of wave energy converter farms. ECCOMAS MARINE Conference, Madrid, Spain - Jun 2023.
- A coupling methodology implementing high-fidelity and reduced-order models for the simulation of bi-fluid flows. ECCOMAS COUPLED PROBLEMS Conference, Chania, Crete, Greece - Jun 2023.
- Model order reduction for wave energy converter farms. SIAM CSE Conference, Amsterdam, Netherlands - Mar 2023.
- Multi-fidelity modeling of wave energy converter farms. RENEW Conference, Lisbon, Portugal - Nov 2022.
- Multi-fidelity multi-scale numerical modeling of wave energy converter farms. Closure of STSM Project at Ghent University, Gent, Belgium - Sep 2022.
- Wave energy converter farms: Modeling strategies. HYWEC Workshop, Bilbao, Spain - Jun 2022.
- Model order reduction for wave energy converter farms. 7th Wave Energy Workshop, Turin, Italy - Apr 2022.

POSTERS

- CFD-ROM coupling technique for the numerical simulation of wave energy converter farms. Numerical Aspects of Hyperbolic Balance Laws and Related Problems – Young Researchers Conference, Ferrara, Italy - Dec 2024.
- Multi-fidelity multi-scale numerical modeling of wave energy converter farms. Workshop on "Reduced-order models at work: Industry and Medicine", Bordeaux, France - Mar 2023.
- Coupling methodologies to enable more effective numerical simulations of WEC farms. 4th WECA Net Assembly, Ghent, Belgium - Mar 2023.
- CFD-ROM coupling technique for the numerical simulation of wave energy converter farms. INORE Symposium 2022, Zarautz, Spain - Oct 2022.
- Multi-fidelity multi-scale numerical modeling of wave energy converter farms. Workshop on "Reduced-order models at work: Industry and Medicine", Bordeaux, France - Mar 2022.

OUTREACH

- Simulating complex real-world phenomena: can we make it affordable? Scientific presentation at the EDMI Doctoral School Day in Mathematics and Computer Science, Bordeaux, France, Apr 2026.
- Participation in the mathematics booth for the science festival "Fête de la Science", Chambéry, France - Oct 2025.
- Participation to the speed meeting with high school young women during the "Filles, maths et informatique : une équation lumineuse" day, Chambéry, France - May 2025.
- Vague océanique : espoir d'une énergie renouvelable ou crainte de la montée des eaux ? Amphi Pour Tous Conference, Chambéry, France - Oct 2024.
- Numerical modeling of wave energy converter farms. Seminar for the Lambda Team (association of the Bordeaux PhD students in Mathematics), Bordeaux, France - May 2023.
- The path to the PhD and an introduction to the PhD project, Talk with Master's students at ENSEIRB-MATMECA, Bordeaux, France, Nov 2022.

LANGUAGES

- **Italian** Native language.
- **French** Fluent. DALF C2 (2018).
- **English** Fluent. TOEFL iBT : 100 (2018). GRE General Test : 156-160-3.5 (2018). TOEIC (2017).
- **Spanish** Basic communication skills.

PROGRAMMING SKILLS

Advanced

Fortran
Matlab
Python

Intermediate

Java
Julia

Basic

C/C++
HTML/CSS

INTERESTS



Environment



Climbing



Mountains



Paragliding



Reading



Cycling

Annex 1.

PUBLICATIONS

Referred Publications in International Journals

- Battisti B., Boscheri W. (2025). *A linearly implicit shock capturing scheme for compressible two-phase flows at all Mach numbers*. Journal of Computational Physics (JCP), 539:114227. DOI
- Battisti B., Bracco G., Bergmann M. (2024). *A multi-fidelity model for wave energy converters*. International Journal for Numerical Methods in Fluids (IJNMF), 97:427-445. DOI
- Battisti B., Giorgi G., Verao Fernandez G. (2024). *Balancing power production and coastal protection: A bi-objective analysis of Wave Energy Converters*. Renewable Energy, 220:119702. DOI
- Cervelli G., Battisti B., Mattiazzo G. (2022). *On the influence of multidirectional irregular waves on the PeWEC device*. Frontiers in Energy Research, 20:908529. DOI

Conference Proceedings Publications

- Boscheri W., Battisti B., *A penalized IMEX finite volume scheme for the isentropic Euler system at all Mach numbers*. Proceedings of the HYP 2024 conference, accepted, 2025. PDF
- Battisti B., Giorgi G., Verao Fernandez G. (2024). *Enhancing synergy for power generation and coastal protection through wave energy converters*. Innovations in Renewable Energies Offshore - RENEW, Lisbon. PDF
- Battisti B., Giorgi G., Verao Fernandez G., Troch P. (2023). *Multi-query analysis of a PeWEC farm*. Proceedings of the European Wave and Tidal Energy Conference - EWTEC, vol. 15, Bilbao. PDF
- Battisti B., Blickhan T., Enchery G., Ehrlacher V., Lombardi D., Mula O. (2023). *Wasserstein model reduction approach for parametrized flow problems in porous media*. ESAIM: Proceedings and Surveys, vol. 73, p. 28-47, Luminy. DOI
- Battisti B., Bracco G., Bergmann M. (2022). *Multi-fidelity modeling of wave energy converter farms*. Trends in Renewable Energies Offshore - RENEW, p. 351-357, Lisbon. website
- Niosi F., Battisti B., Sirigu S.A. (2022). *Influence of hydrodynamic interactions on the productivity of PeWEC wave energy converter array*. International Conference on Electrical, Computer, Communications and Mechatronics Engineering - ICECCME, Maldives. PDF
- Casalone P., Dell'Edera O., Fontana M., Battisti B., Mattiazzo G. (2022). *Solutions to Wave Damping Over Time in CFD RANS Simulations Due to Exponential Generation of Numerical Turbulence*. Proceedings of the International Conference on Offshore Mechanics and Arctic Engineering - OMAE, vol. 8, Hamburg. website
- Fontana M., Casalone P., Dell'Edera O., Niosi F., Battisti B., Mattiazzo G. (2021). *Viscous damping analysis of WEC's hull in yaw motion - Methodology and viscous damping parameters identification*. Proceedings of the European Wave and Tidal Energy Conference - EWTEC, Plymouth. website